

Table 20: An example from the dataset for each type of implicature found in the test set. The rightmost column shows the amount of that type we manually found in the test set.

Type	Example Utterance	Example Response	Impl.	#
Generalised	You know all these people?	Some.	No.	47
Particularised	Want to stay for a nightcap?	I’ve gotta get up early.	No.	94
World knowledge	Did you leave fingerprints?	I wore gloves.	No.	23
Idiom	Would he fire me?	He’s all bark and no bite.	No.	42
Rhetorical question	Can you drive that far?	Can fish swim?	Yes.	11
Other	-	-	-	383

K.9 Detailed results type label analysis

In the main paper we do an analysis of two types of examples that occur frequently in the dataset, namely generalised and particularised implicatures. Here, we detail the full taxonomy of types of examples occurring in the dataset and report detailed results for each type per model (see 21 until Table 34 below). In Table 20 the full taxonomy of the examples is shown, representing types of examples that occur frequently in the dataset. We manually labeled 217 examples of the 600 examples in the test set according to this taxonomy. The remaining 383 examples do not fall as clearly within a category and are grouped together as type *other*. *Generalised* implicatures require little or no context to be understood. They are the simplest type of example in the test set, and generally imply the same thing (“some” almost always implies “not all”). *Particularised* implicatures, by contrast, do require context to be resolved. For example, from Table 20, we need the context that it is undesirable to stay up late drinking when one has to get up early (see in Appendix D more on generalised vs. particularised). The type *world knowledge* requires knowledge of the physical world to be resolved. From the example in Table 20; we need to know that you cannot leave fingerprints when wearing gloves to resolve this implicature. *Idiom* types contain an idiom or a metaphor that one needs to know or understand to resolve the implicature, and finally *Rhetorical question* types contain a question like “Is the Pope Catholic?”, often requiring factual knowledge to be resolved.

The following tables contain the detailed results broken down per example type: Table 21 - Table 34. The most interesting pattern in this data is that for almost all models, even the best model (GPT-4 30-shot in Table 32), there is a significant gap between human-level performance on the particularised examples. This gap is larger than the gap for the other labels usually. Few-shot prompting can often mitigate this (e.g. for GPT-3-175b, Cohere-52b, and text-davinci-002), but not always (e.g. for GPT-4 the gap remains large for $k = 30$). However, for GPT-4, chain-of-thought can mitigate the gap as seen in Table 34. Where GPT-4 30-shot obtains 71.97% accuracy on the particularised examples (and humans 83.18%), GPT-4 with 5-shot CoT achieves 81.63%, which is close to human-level. We find that the particularised examples mostly benefit from CoT prompting. Namely, for the generalised type of examples, GPT-4 30-shot already achieves 86.23% accuracy and CoT improves this to 88.66%, which is a much smaller improvement than for the particularised examples.

K.10 Detailed results per model

This section contains the results used for the zero-shot and few-shot evaluation in the main text in Section 4, broken down per prompt template. See Table 35 until Table 84.

Table 21: Accuracy per label for 0-shot evaluation.

Model	Mean	World knowledge	Idiom	Rhetorical question
OPT-125m	50.92	50.00 +/- 2.17	51.52 +/- 9.96	57.58 +/- 10.05
OPT-350m	57.14	57.97 +/- 10.25	64.77 +/- 3.65	65.15 +/- 3.39
OPT-1.3b	60.36	60.14 +/- 5.84	68.94 +/- 5.52	59.09 +/- 4.55
OPT-2.7b	59.56	60.87 +/- 6.15	67.05 +/- 2.18	69.70 +/- 6.78
OPT-6.7b	60.33	59.42 +/- 6.95	59.47 +/- 2.04	53.03 +/- 19.93
OPT-13b	61.03	63.77 +/- 14.78	73.86 +/- 7.51	66.67 +/- 16.32
OPT-30b	61.47	65.94 +/- 10.48	62.88 +/- 8.05	74.24 +/- 6.25
OPT-66b	61.33	69.57 +/- 13.75	60.23 +/- 4.30	59.09 +/- 18.74
OPT-175b	55.33	55.07 +/- 5.42	54.55 +/- 9.19	63.64 +/- 21.64
BLOOM-560m	51.58	54.35 +/- 5.47	54.92 +/- 16.72	50.00 +/- 13.64
BLOOM-1b1	51.17	50.00 +/- 2.17	50.38 +/- 11.77	53.03 +/- 12.22
BLOOM-1b7	53.61	52.17 +/- 6.15	53.79 +/- 8.77	68.18 +/- 6.94
BLOOM-3b	56.89	54.35 +/- 6.02	59.85 +/- 4.48	63.64 +/- 5.25
BLOOM-7b1	58.67	63.77 +/- 14.57	68.94 +/- 5.82	68.18 +/- 4.55
BLOOM-176b	54.22	55.07 +/- 7.39	50.38 +/- 11.01	62.12 +/- 9.70
EleutherAI-125m	51.89	56.52 +/- 9.72	52.65 +/- 8.84	63.64 +/- 5.25
EleutherAI-1.3b	53.14	51.45 +/- 3.90	53.03 +/- 11.19	62.12 +/- 3.39
EleutherAI-2.7b	59.17	60.14 +/- 13.38	65.91 +/- 3.94	68.18 +/- 4.55
EleutherAI-6b	56.36	57.25 +/- 7.28	56.06 +/- 8.87	50.00 +/- 17.99
EleutherAI-20b	57.53	51.45 +/- 3.90	67.80 +/- 5.93	72.73 +/- 5.25
Cohere-409m	51.61	52.17 +/- 4.35	53.41 +/- 11.94	54.55 +/- 12.86
Cohere-6b	57.28	55.80 +/- 5.28	60.23 +/- 5.98	72.73 +/- 9.09
Cohere-13b	57.19	59.42 +/- 4.81	54.55 +/- 10.82	48.48 +/- 10.05
Cohere-52b	58.50	60.14 +/- 13.61	65.15 +/- 3.86	74.24 +/- 11.03
GPT-3-350m	51.47	51.45 +/- 3.90	53.41 +/- 13.56	50.00 +/- 13.64
GPT-3-1.3b	57.72	61.59 +/- 11.06	64.39 +/- 4.08	65.15 +/- 3.39
GPT-3-6.7b	54.83	54.35 +/- 6.99	53.79 +/- 7.61	62.12 +/- 3.39
GPT-3-175b	57.22	55.80 +/- 7.28	68.94 +/- 5.19	77.27 +/- 8.70
T0-3b	48.25	54.35 +/- 4.86	42.42 +/- 4.29	36.36 +/- 0.00
T0-11b	55.61	60.14 +/- 6.84	54.92 +/- 14.93	36.36 +/- 0.00
BlenderBot-90m	46.64	52.17 +/- 0.00	38.64 +/- 0.00	36.36 +/- 0.00
BlenderBot-3b	53.44	47.83 +/- 0.00	61.36 +/- 1.31	63.64 +/- 0.00
BlenderBot-9b	53.36	52.17 +/- 6.64	60.98 +/- 4.81	63.64 +/- 0.00
Flan-T5-780m	63.31	72.46 +/- 4.10	71.97 +/- 5.82	54.55 +/- 13.89
Flan-T5-3b	52.50	50.72 +/- 6.95	51.89 +/- 4.23	42.42 +/- 8.57
Flan-T5-11b	60.78	65.94 +/- 5.84	72.35 +/- 7.59	65.15 +/- 6.25
Cohere-command-6b	66.31	72.46 +/- 7.80	78.41 +/- 4.30	37.88 +/- 3.39
Cohere-command-52b	60.22	66.67 +/- 10.85	63.64 +/- 10.33	77.27 +/- 6.94
text-ada-001-unknown	56.50	63.77 +/- 4.10	58.71 +/- 16.04	51.52 +/- 10.05
text-babbage-001-unknown	64.47	67.39 +/- 6.02	76.52 +/- 1.69	60.61 +/- 10.05
text-curie-001-unknown	68.94	76.81 +/- 3.24	76.89 +/- 2.76	54.55 +/- 12.86
text-davinci-001-unknown	72.31	84.78 +/- 7.43	78.79 +/- 4.08	59.09 +/- 13.64
text-davinci-002-unknown	70.58	82.61 +/- 9.05	75.38 +/- 3.05	57.58 +/- 16.32
text-davinci-003-unknown	71.25	86.96 +/- 13.28	72.35 +/- 7.35	48.48 +/- 8.57
ChatGPT-unknown	72.08	82.61 +/- 12.04	83.33 +/- 5.97	56.06 +/- 16.11
GPT-4-unknown	81.78	92.03 +/- 2.99	90.91 +/- 3.21	84.85 +/- 8.57
Humans	86.23	93.04	92.73	92.73

Table 22: Accuracy per label for 0-shot evaluation.

Model	Mean	Particularised	Generalised	Other
OPT-125m	50.92	49.43 +/- 5.52	55.07 +/- 21.10	50.56 +/- 1.33
OPT-350m	57.14	47.92 +/- 4.37	69.20 +/- 8.36	56.68 +/- 4.99
OPT-1.3b	60.36	51.52 +/- 6.81	74.64 +/- 2.05	59.65 +/- 3.51
OPT-2.7b	59.56	50.19 +/- 5.06	69.93 +/- 5.53	59.22 +/- 6.14
OPT-6.7b	60.33	52.27 +/- 5.90	75.36 +/- 2.71	60.77 +/- 6.13
OPT-13b	61.03	55.49 +/- 8.79	75.00 +/- 5.72	58.79 +/- 5.51
OPT-30b	61.47	54.55 +/- 4.15	71.38 +/- 5.94	61.11 +/- 2.12
OPT-66b	61.33	55.11 +/- 7.33	69.93 +/- 12.26	61.46 +/- 3.68
OPT-175b	55.33	54.17 +/- 7.70	58.33 +/- 18.51	55.12 +/- 4.21
BLOOM-560m	51.58	50.76 +/- 5.59	48.91 +/- 25.22	51.59 +/- 3.50
BLOOM-1b1	51.17	50.57 +/- 6.32	53.26 +/- 27.40	51.16 +/- 2.41
BLOOM-1b7	53.61	50.38 +/- 7.95	59.78 +/- 18.82	53.23 +/- 1.62
BLOOM-3b	56.89	51.70 +/- 8.27	67.39 +/- 10.35	56.46 +/- 4.39
BLOOM-7b1	58.67	46.59 +/- 2.86	79.35 +/- 2.74	57.11 +/- 4.03
BLOOM-176b	54.22	54.73 +/- 10.61	60.14 +/- 16.73	53.57 +/- 1.65
EleutherAI-125m	51.89	50.38 +/- 5.71	57.25 +/- 20.15	50.90 +/- 0.99
EleutherAI-1.3b	53.14	50.57 +/- 7.03	55.43 +/- 23.20	53.32 +/- 2.05
EleutherAI-2.7b	59.17	50.95 +/- 7.43	74.64 +/- 4.97	58.10 +/- 2.92
EleutherAI-6b	56.36	53.22 +/- 6.86	69.20 +/- 7.36	55.73 +/- 2.43
EleutherAI-20b	57.53	49.43 +/- 6.68	72.83 +/- 5.99	56.20 +/- 3.01
Cohere-409m	51.61	51.33 +/- 4.84	52.54 +/- 22.71	51.25 +/- 2.93
Cohere-6b	57.28	51.52 +/- 6.68	64.49 +/- 16.06	57.06 +/- 3.13
Cohere-13b	57.19	52.27 +/- 5.83	68.12 +/- 10.99	57.45 +/- 3.35
Cohere-52b	58.50	51.52 +/- 7.21	73.91 +/- 5.75	56.85 +/- 3.81
GPT-3-350m	51.47	50.76 +/- 6.96	52.90 +/- 24.07	51.29 +/- 1.63
GPT-3-1.3b	57.72	50.00 +/- 6.29	67.75 +/- 9.84	57.06 +/- 3.78
GPT-3-6.7b	54.83	52.65 +/- 8.16	63.41 +/- 15.14	54.26 +/- 2.12
GPT-3-175b	57.22	53.03 +/- 1.93	71.01 +/- 4.81	54.61 +/- 5.58
T0-3b	48.25	55.68 +/- 0.66	27.17 +/- 2.74	49.83 +/- 1.90
T0-11b	55.61	57.95 +/- 2.18	47.10 +/- 17.15	56.33 +/- 6.49
BlenderBot-90m	46.64	55.49 +/- 0.42	23.91 +/- 0.00	48.32 +/- 0.00
BlenderBot-3b	53.44	44.51 +/- 0.42	76.09 +/- 0.00	51.81 +/- 0.20
BlenderBot-9b	53.36	49.24 +/- 4.81	71.01 +/- 5.71	51.03 +/- 1.63
Flan-T5-780m	63.31	59.28 +/- 3.90	68.84 +/- 7.60	62.23 +/- 3.13
Flan-T5-3b	52.50	54.55 +/- 1.61	48.19 +/- 11.73	53.14 +/- 3.15
Flan-T5-11b	60.78	51.52 +/- 3.39	73.19 +/- 7.28	59.60 +/- 2.18
Cohere-command-6b	66.31	58.90 +/- 3.62	73.19 +/- 2.71	66.15 +/- 2.41
Cohere-command-52b	60.22	55.49 +/- 4.66	60.51 +/- 16.67	59.99 +/- 5.09
text-ada-001-unknown	56.50	52.65 +/- 3.86	61.59 +/- 15.96	56.24 +/- 5.74
text-babbage-001-unknown	64.47	56.25 +/- 2.52	72.46 +/- 9.86	63.87 +/- 1.55
text-curie-001-unknown	68.94	66.48 +/- 2.34	68.84 +/- 5.98	68.48 +/- 3.63
text-davinci-001-unknown	72.31	59.66 +/- 5.07	79.35 +/- 9.78	73.17 +/- 2.54
text-davinci-002-unknown	70.58	64.20 +/- 3.75	80.07 +/- 5.67	69.94 +/- 3.69
text-davinci-003-unknown	71.25	63.64 +/- 1.86	82.25 +/- 4.77	71.23 +/- 2.74
ChatGPT-unknown	72.08	68.75 +/- 2.99	69.57 +/- 11.16	71.66 +/- 5.79
GPT-4-unknown	81.78	71.59 +/- 3.47	89.86 +/- 2.05	81.35 +/- 1.66
Humans	86.23	83.18	92.17	84.86